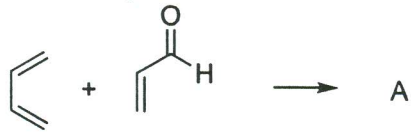
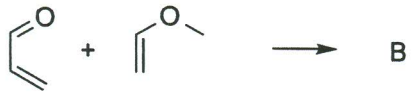
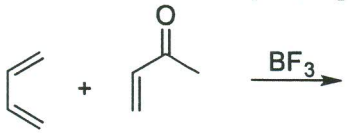
Organic Cume 12-5-2014

1. According to the FMO (Frontier Molecular Orbital) theory, draw the HOMO (highest occupied molecular orbital) and the LUMO (lowest occupied molecular orbital) of the following Diels-Alder reactions that have the most important interactions. Give the structure of the products **A** and **B**.

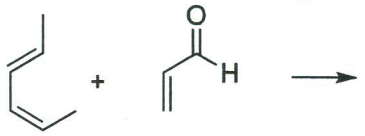
1) 

2) 

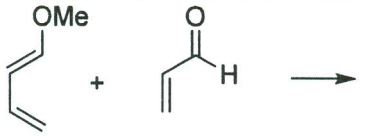
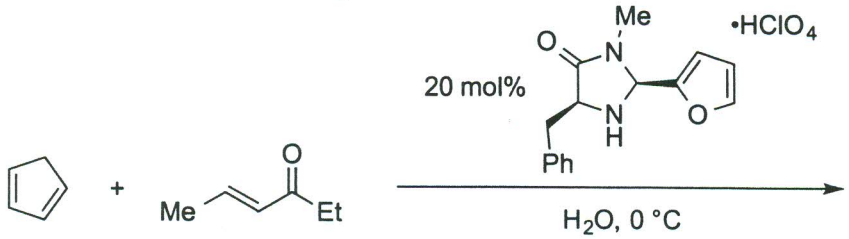
2. Lewis acids are known to accelerate the Diels-Alder reactions. Using the following reaction as an example, explain the rate acceleration based on the FMO theory.



3. The endo selectivity is often observed in the Diels-Alder reactions. Draw the endo product of the following reaction. Use the FMO theory to explain the endo selectivity.



4. The regioselectivity may arise when unsymmetrical dienes and dienophiles are used in the Diels-Alder reactions. Give the major regioisomer of the following reaction (you can ignore the endo/exo selectivity and the facial selectivity with respect to the diene and the dienophile). Use resonance structures to explain the regioselectivity.



5. Give the major product with stereochemistry of the following Diels-Alder reaction. Rationalize the sense of asymmetric induction for this reaction.